

INDUSTRIAL TRAINING PROJECT

Smart Paper Mill Production Monitoring Dashboard

A real-time, role-based web dashboard for monitoring paper production
Industrial training carried out at SATIA INDUSTRIES LIMITED

Submitted by

Manreet Gill (UID: 24BCS10479)

B.E. Computer Science & Engineering - Chandigarh University

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Organisation & Domain

- **Satia Industries Ltd.** - a leading agro-residue paper manufacturer producing writing & printing paper primarily from **wheat straw** and wood pulp.
- Production flows through several stages: **pulping** -> **stock preparation** -> **paper machine (PM)** -> **finishing**, run on multiple paper machines.
- Key process parameters: output (tonnes), **GSM**, **moisture**, machine speed, temperature, running hours and efficiency.
- Continuous, round-the-clock operation across **three shifts (A / B / C)** makes timely visibility of production, quality and machine health business-critical.
- Disclaimer: all figures in this project are simulated and industry-realistic - no confidential Satia data is used.

Problem Statement

- Production is recorded **manually** in shift registers, logbooks and disconnected spreadsheets kept separately for each section.
- **No consolidated, real-time view** of output, quality (GSM / moisture), machine status, downtime or efficiency across the plant.
- Out-of-spec quality and excessive downtime are often **detected late**, after the shift, when corrective action is harder.
- **KPIs such as OEE are tedious to compute** by hand, and management reports take time to compile from scattered sources.
- Data is hard to secure, audit or share role-by-role across the organisation.

Objectives

- Provide a **centralised, role-based web dashboard** for real-time production monitoring.
- Capture production output and **quality parameters (GSM, moisture, speed)** plus **downtime events**, persisted in a database.
- Show **live machine status and telemetry** and raise **threshold-driven alerts** automatically.
- Compute **OEE and key KPIs** and let users **export reports (CSV / PDF)**.
- Enforce **secure authentication and access control**, and run on standard, low-cost student hardware and a normal browser.

Scope

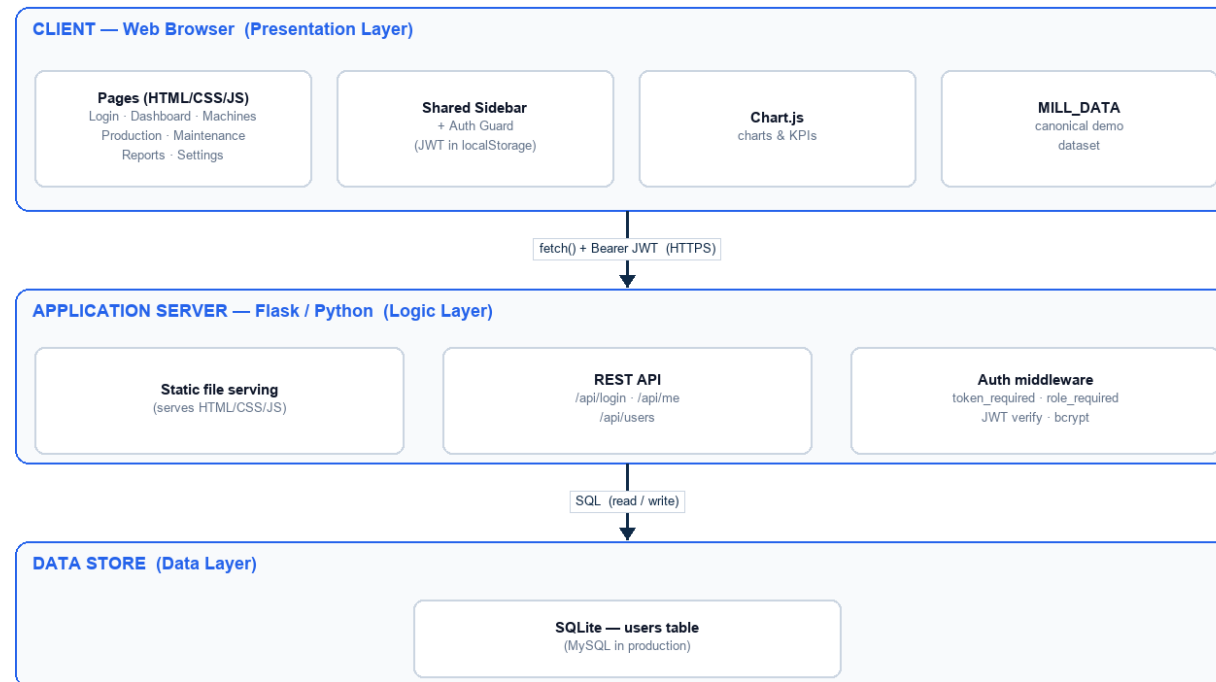
- **In scope:** a complete web application - responsive front-end, REST API back-end and a database - with four user roles and nine functional modules.
- **In scope:** authentication, production & quality entry, machine monitoring, maintenance, downtime logging, reports, alerts, settings and an admin panel.
- **Simulated sensors:** live machine telemetry is streamed from the server to mimic a real sensor feed, since plant PLC/DCS hardware is not available to a student project.
- **Out of scope:** direct PLC / DCS / IoT sensor integration, ERP/SAP integration, and a native mobile application - identified instead as future scope.

Technology Stack

Layer	Technology	Why
Front-end	HTML5, CSS3, JavaScript, Chart.js	Lightweight, no build step, runs anywhere
Back-end	Python - Flask (REST API + static)	Simple, same-origin API & UI
Database	SQLite (MySQL is production target)	Zero-config, file-based persistence
Security	bcrypt, PyJWT	Hashed passwords + stateless JWT auth
Real-time	Server-Sent Events (SSE)	Live push with zero extra dependencies
Testing	pytest (94 automated tests)	Reproducible backend verification
Docs / Deploy	python-docx, Matplotlib; gunicorn / Render	Auto-generated docs; free hosting

System Architecture

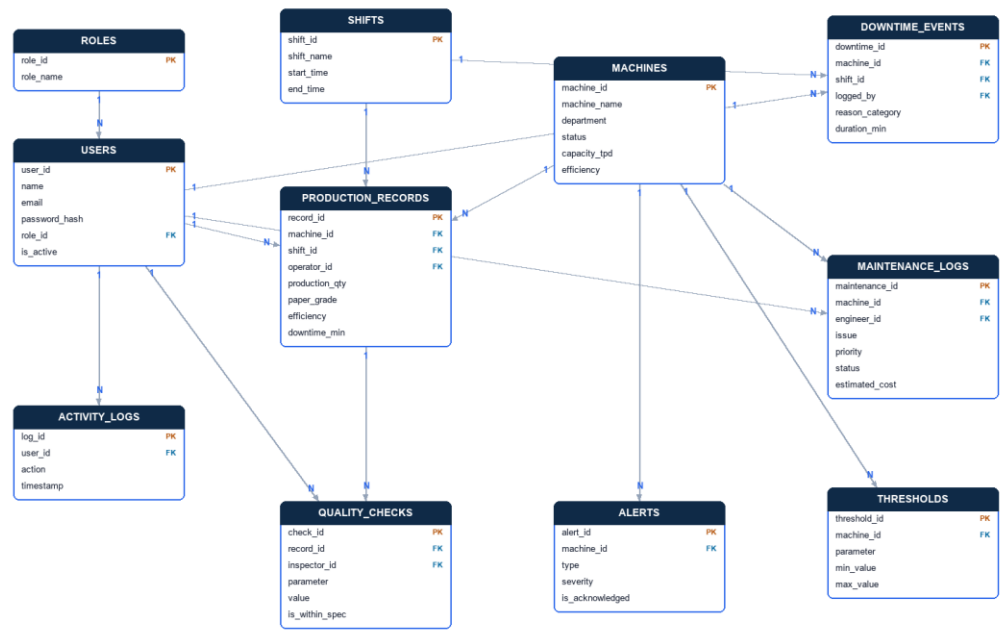
System Architecture — Smart Paper Mill Production Monitoring Dashboard



Three-tier architecture: browser front-end -> Flask REST API + SSE -> SQLite database (same origin).

Database Design

Entity-Relationship Diagram — Smart Paper Mill Production Monitoring Dashboard



PK = Primary Key FK = Foreign Key 1 — N = one-to-many relationship

Core tables: users, production_records, maintenance_logs, downtime_events, machines, thresholds, activity_logs.

User Roles & Access Control

Role	Representative permissions
Administrator	Full access; manage users, machines & thresholds; view audit log
Plant Manager	View all dashboards & reports; oversee production, OEE and downtime
Shift Supervisor	Log production, downtime and maintenance for their shift
Machine Operator	Enter production & quality readings; view machine status & alerts

Modules & Key Features

- **Dashboard** - KPI cards (incl. average OEE), production-trend & machine-status charts, live feed.
- **Production** - record output and **GSM / moisture / speed**; out-of-spec values flagged.
- **Machines** - colour-coded status board (Running / Idle / Stopped / Maintenance) with live temperature & efficiency.
- **Maintenance & Downtime** - log and track jobs by priority/status; capture stoppages with auto-computed duration.
- **Reports, Alerts, Settings & Admin** - filtered reports with CSV/PDF export, threshold alerts, dark mode, and user/machine/threshold management.

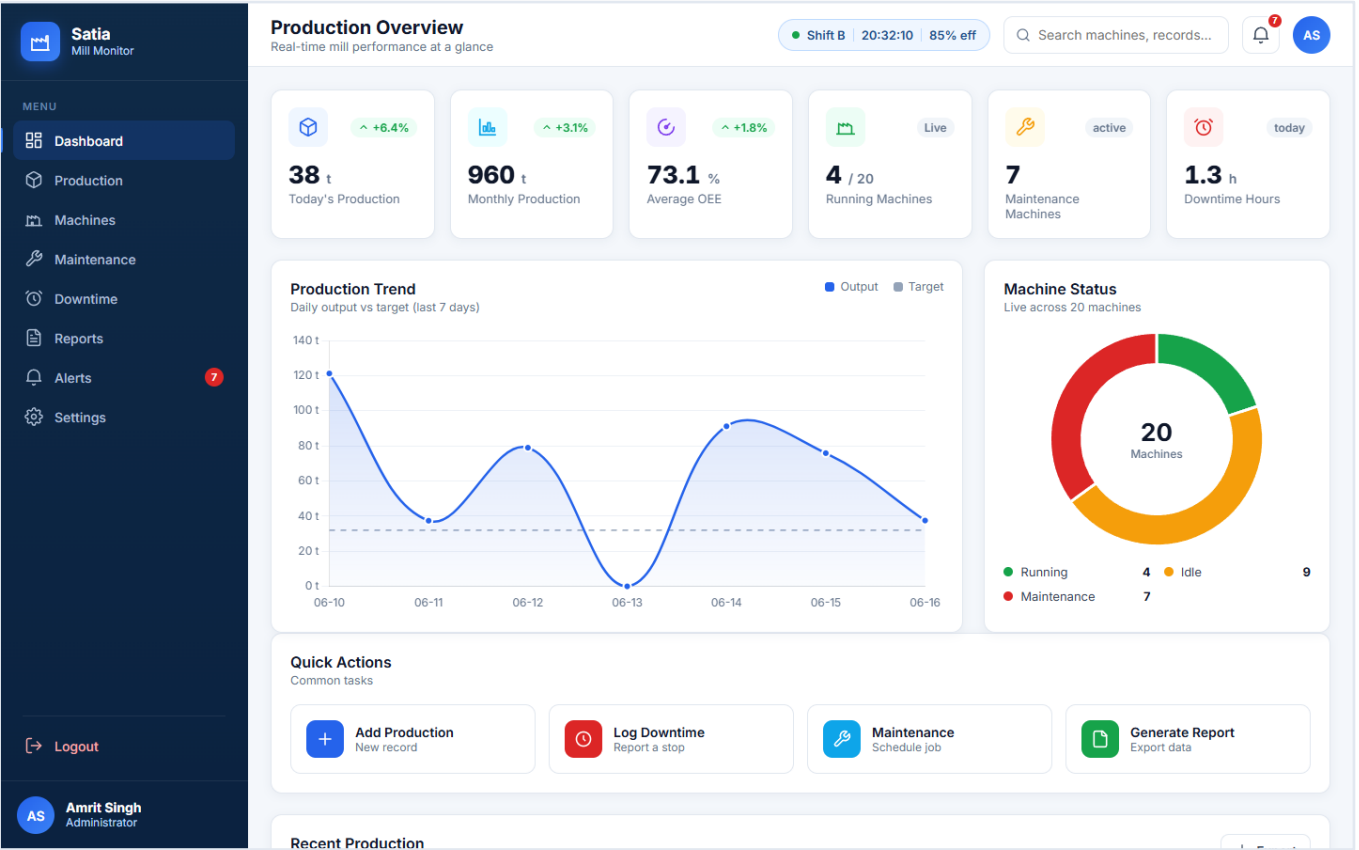
Real-Time Monitoring

- The server **pushes live readings** (temperature, efficiency) to every open page using **Server-Sent Events** over an authenticated stream.
- Machine cards and the dashboard's live indicator **update every few seconds** without reloading the page; a 'Live' timestamp shows freshness.
- **KPIs, reports and alerts are API-driven:** records entered through the app immediately feed the dashboard KPIs, the daily report and the alerts list.
- Pages **degrade gracefully** - if the backend is unavailable they fall back to a seeded demonstration dataset, so the UI never breaks.

Quality, OEE & Alerts

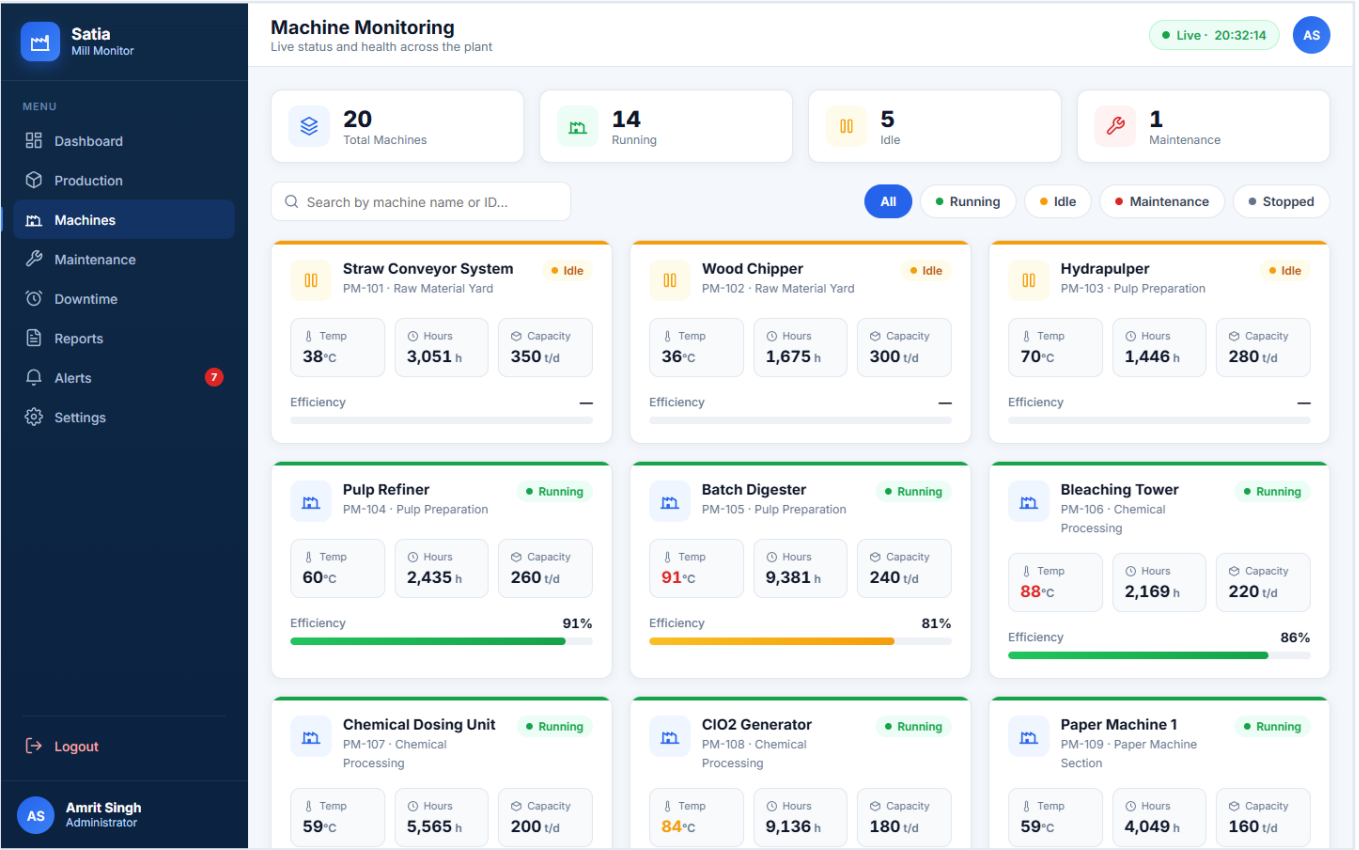
- Every production record carries **quality readings - GSM, moisture and machine speed** - validated against configurable limits.
- **Thresholds:** GSM 45-120, moisture $\leq 6.5\%$, temperature ≤ 88 deg C, efficiency $\geq 75\%$, downtime ≤ 60 min (all admin-editable).
- **OEE = Availability x Performance x Quality** is computed and charted per machine.
- **Threshold-driven alerts** are raised automatically for out-of-spec GSM/moisture and over-limit downtime, with acknowledge & severity filtering.

Dashboard



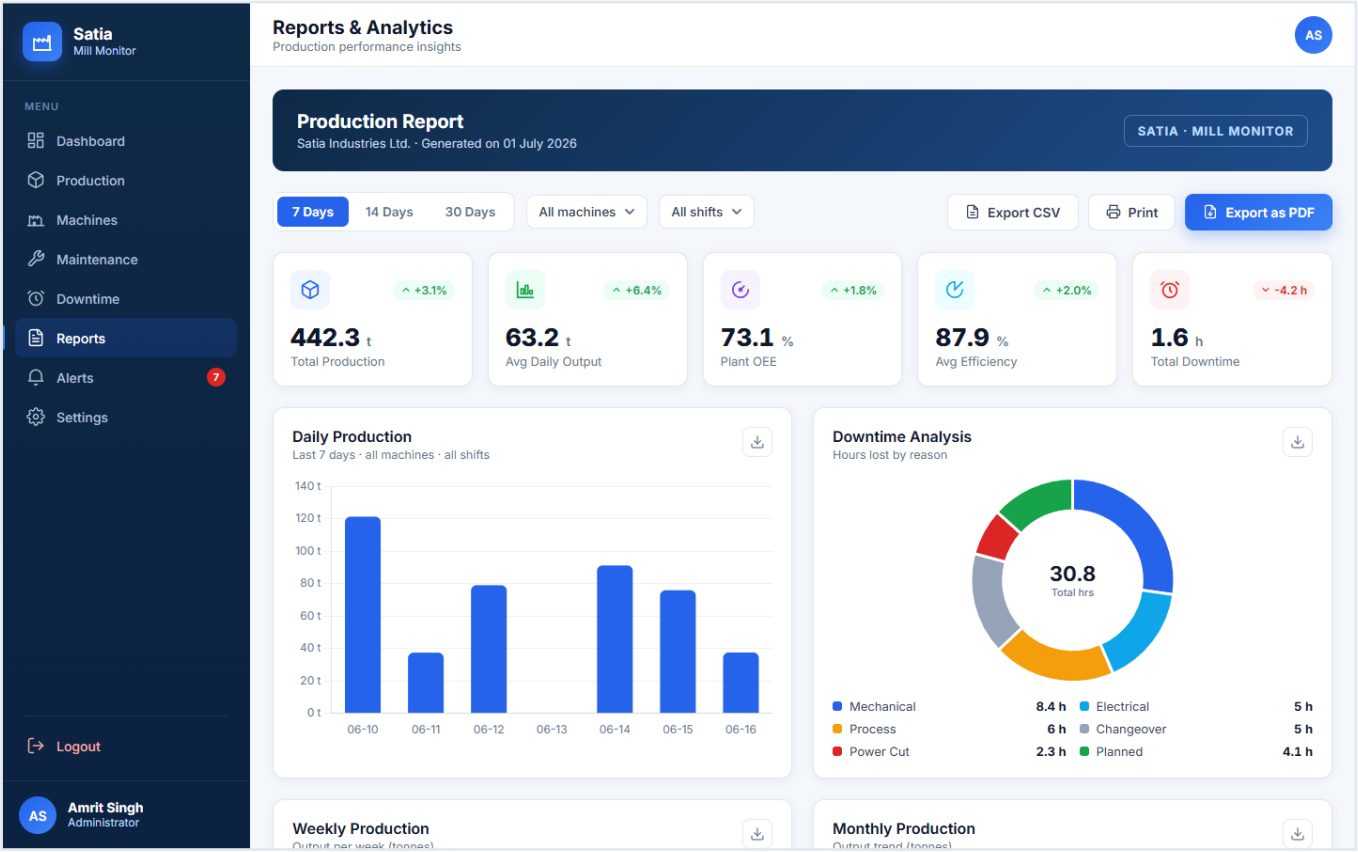
Live production overview: KPI cards, production-trend and machine-status charts, and quick actions.

Machine Monitoring



Colour-coded status board - Running / Idle / Stopped / Maintenance - with live temperature & efficiency.

Reports & Analytics



Period / machine / shift filters, plant OEE, downtime breakdown, and CSV / PDF export.

Production Entry & Alerts

Satia
Mill Monitor

MENU

- Dashboard
- Production
- Machines
- Maintenance
- Downtime
- Reports
- Alerts
- Settings

Logout

AS
Amrit Singh
Administrator

Production Entry

Record and manage shift-wise production data

Add Production Record

Fill in the details and save the entry

Date *

07/01/2026

Shift *

Select shift

Machine *

Select machine

Operator *

Select operator

Production Quantity (tonnes) *

e.g. 18.5

Paper Grade *

Select grade

GSM (g/m²)

e.g. 70

Moisture (%)

e.g. 5.2

Machine Speed (m/min)

e.g. 620

Remarks

Optional notes (e.g. sheet break, speed change)...

Save Record

Reset

Cancel

Production Records

24 record(s) found

Q Search records...

DATE	SHIFT	MACHINE	OPERATOR	QTY (T)	GRADE	GSM	MOISTURE %	SPEED	REMARKS	ACTIONS
2026-06-02	Shift Night	PM-110	Amrit Singh	43.9	Copier Paper (70 GSM)	120	6.1	625	Minor speed dip	
2026-06-02	Shift Night	PM-110	Harpreet Kaur	39.5	Premium Bond Paper	80	5.2	620	Minor speed dip	

Production entry with GSM / moisture / speed quality capture.

Satia
Mill Monitor

MENU

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Logout

AS
Amrit Singh
Administrator

Alerts & Notifications

Threshold and machine-status alerts

14
Total Alerts

14
Unacknowledged

0
Critical

14
Warnings

All

Unacknowledged

Critical

Warning

Info

Mark all read

Extended downtime

WARNING

PM-116: 80 min of downtime (Changeover) exceeds the 60-min threshold.

PM-116 16 Jun 2026

Acknowledge

Extended downtime

WARNING

PM-112: 128 min of downtime (Changeover) exceeds the 60-min threshold.

PM-112 12 Jun 2026

Acknowledge

Extended downtime

WARNING

PM-116: 165 min of downtime (Electrical) exceeds the 60-min threshold.

PM-116 12 Jun 2026

Acknowledge

Extended downtime

WARNING

PM-105: 121 min of downtime (Raw Material) exceeds the 60-min threshold.

PM-105 11 Jun 2026

Acknowledge

Extended downtime

WARNING

PM-120: 64 min of downtime (Changeover) exceeds the 60-min threshold.

PM-120 08 Jun 2026

Acknowledge

Extended downtime

WARNING

PM-105: 141 min of downtime (Changeover) exceeds the 60-min threshold.

PM-105 06 Jun 2026

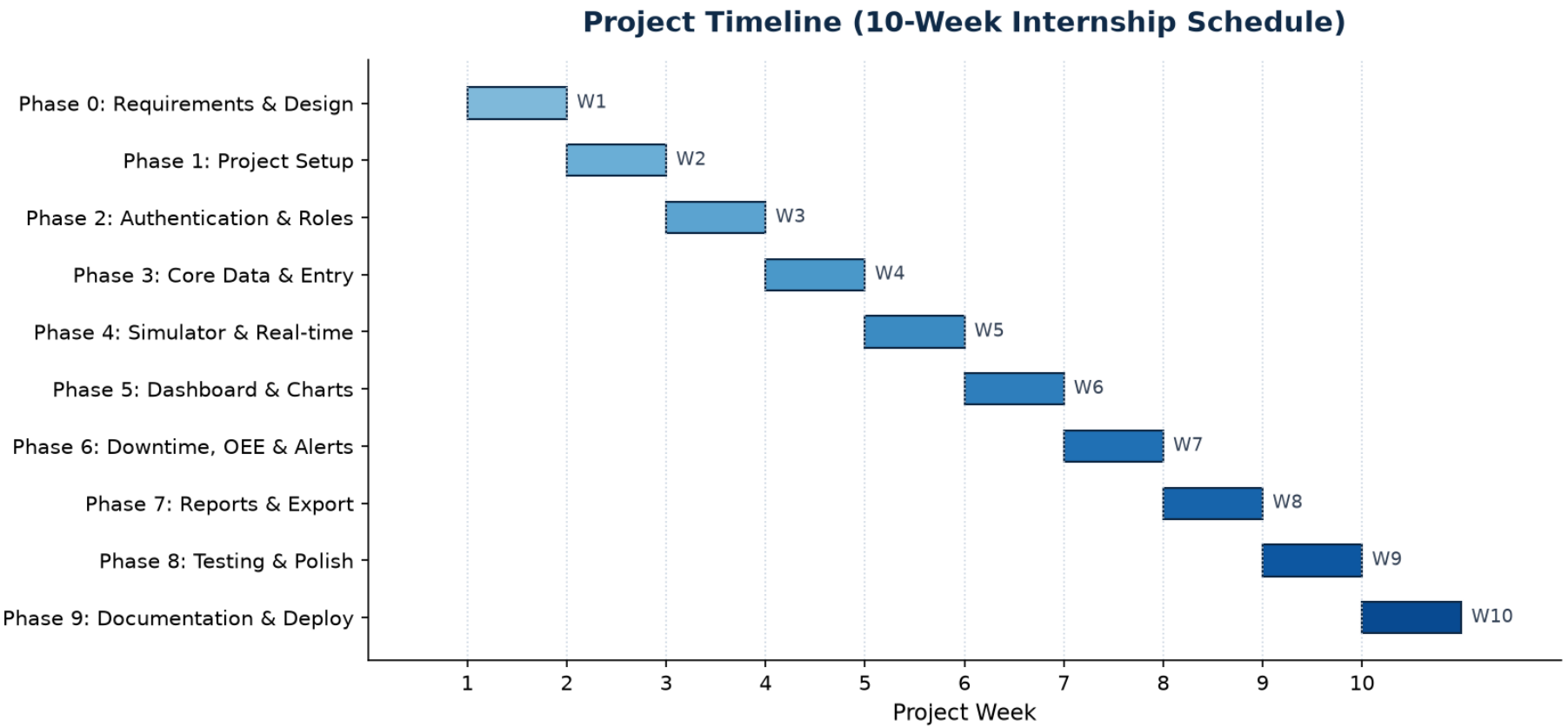
Acknowledge

Threshold-driven alerts with severity filtering and acknowledgement.

Smart Paper Mill Production Monitoring Dashboard - Satia Industries Ltd.

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Project Timeline



Phased delivery: requirements & design -> module build -> backend & security -> testing, docs & deployment.

Testing & Quality Assurance

Test area	Coverage
Authentication & JWT	Valid/invalid login, token guard, expiry handling
Role-based access control	Allowed vs. forbidden endpoints per role (403/200)
Production / Maintenance / Downtime	Full create-read-update-delete + validation
Quality & thresholds	GSM / moisture / speed fields and limit checks
Live stream & audit	Authenticated SSE; audit-log entries
Role workflows	End-to-end tasks for each of the four roles

Results & Outcomes

- A **working, end-to-end web application** with real authentication, role-based access and **database persistence that survives restarts**.
- **Live telemetry** plus API-driven KPIs, reports and alerts across the dashboard.
- A **canonical, seeded dataset** (machines, production, maintenance, operators) for a consistent, repeatable demonstration.
- **94/94 automated tests passing**, and a complete documentation set - **SRS, project report and role-based user manuals**.
- **Deployment-ready** for free hosting (Render) via gunicorn, with no paid services.

Challenges & Learnings

- **Simulating real-time data without hardware** - solved cleanly with Server-Sent Events instead of polling or heavyweight sockets.
- **Keeping documentation consistent with code** - a single source of truth (canonical dataset + build scripts) prevented drift.
- **Security details** - e.g. correct JWT subject typing and enforcing role checks on the server, not just the UI.
- **Designing for constraints** - the whole stack had to run on a low-spec laptop and a normal browser, which shaped every technology choice.
- Gained hands-on experience across **full-stack development, REST APIs, auth, testing and deployment.**

Future Scope

- Integrate **real plant sensors via PLC / DCS / IoT gateways** to replace the simulated feed.
- Migrate to **MySQL / PostgreSQL** for multi-user, production-scale deployment.
- Add **predictive maintenance** using machine-learning on historical machine data.
- Deliver a **mobile app** and **email / SMS alert** notifications for supervisors on the move.
- Extend to **multi-plant / multi-line** monitoring with consolidated group dashboards.

Conclusion

- The project delivers a **secure, real-time, role-based monitoring dashboard** that turns manual, scattered records into a **single live view** of the plant.
- It demonstrates a **complete, tested and documented full-stack system** built entirely with free, open-source tools.
- It directly addresses the industry need for **timely production, quality and machine-health visibility** - and provides a clear path to a sensor-connected production system.

Thank You

Questions & Discussion

Manreet Gill - B.E. CSE, Chandigarh University
Industrial Training Project - Satia Industries Limited